

5.6 Classwork: “Show That” Problems (no calculator)

IB command term: **show that** — obtain the required result without the formality of proof.

“Show that” questions are *not* verify; you may not simply substitute values and show that the equation holds.

Instead, you must derive the given result with all of the algebraic steps, including the final, given result.

1. [Maximum mark: 16]

A particle P moves along the x -axis. The velocity of P is $v \text{ m s}^{-1}$ at time t seconds, where

$$v(t) = 4 + 4t - 3t^2 \quad \text{for } 0 \leq t \leq 3.$$

When $t = 0$, P is at the origin O .

(a) (i) Find the value of t when P reaches its maximum velocity.

(ii) Show that the distance of P from O at this time is $\frac{88}{27}$ metres. [7]

(b) Sketch a graph of v against t , clearly showing any points of intersection with the axes. [4]

(c) Find the total distance travelled by P . [5]

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2. [Maximum mark: 13]

Let $y = \frac{\ln x}{x^4}$ for $x > 0$.

(a) Show that $\frac{dy}{dx} = \frac{1 - 4 \ln x}{x^5}$. [3]

(b) The graph of f has a horizontal tangent at point P . Find the coordinates of P . [5]

(c) Given that $f''(x) = \frac{20 \ln x - 9}{x^6}$, show that P is a local maximum point. [3]

(d) Solve $f(x) > 0$ for $x > 0$. [2]

Adapted from IB AA SL 2021 May P1 TZ1 Q8